

EEMI

INTRODUCTION

MEASUREMENT:

STANDARD:

MEASURING INSTRUMENT:

The device or instrument used for comparing the unknown quantity with the unit of measurement or a standard quantity is called a measuring instrument.

The value of the unknown quantity can be measured by direct or indirect methods.

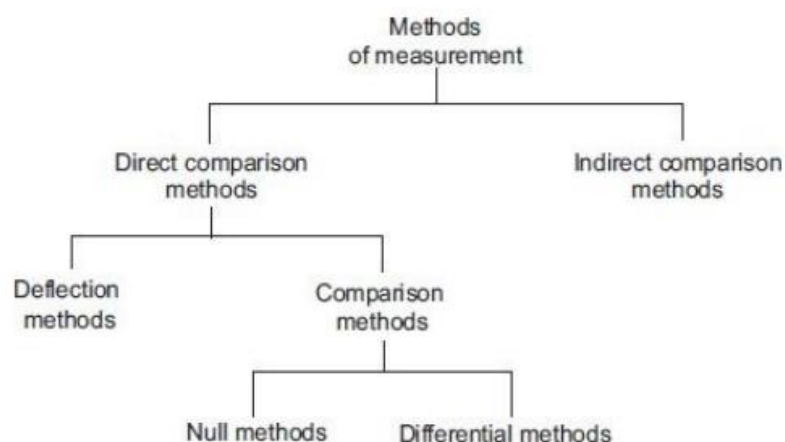
In direct measurement methods, the unknown quantity is measured directly instead of comparing it with a standard. Examples of direct measurement are current by ammeter, voltage by voltmeter, resistance by ohmmeter, power by wattmeter, etc.

In indirect measurement methods, the value of the unknown quantity is determined by measuring the functionally related quantity and calculating the desired quantity rather than measuring it directly.

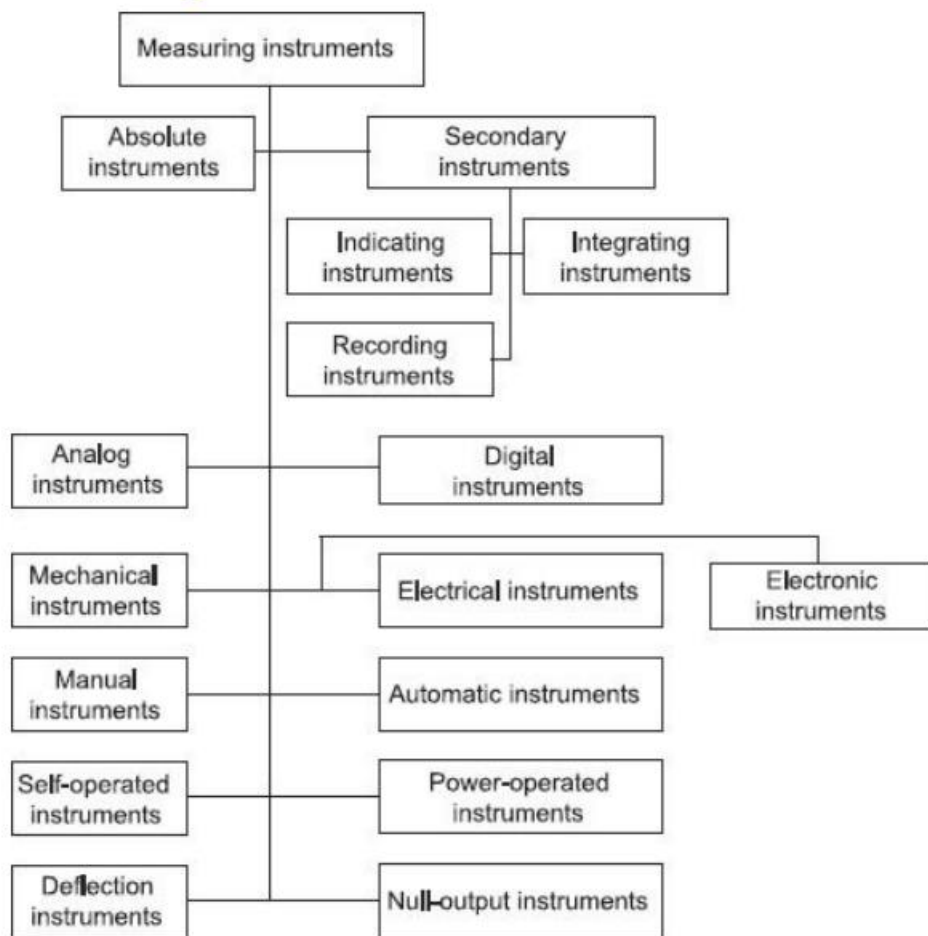
FUNDAMENTAL AND DERIVED UNITS:

METHODS OF MEASUREMENT:

- Direct comparison methods
- Indirect comparison methods



CLASSIFICATION OF INSTRUMENTS:



DEFINITIONS OF SOME STATIC CHARACTERISTICS:

1. Accuracy :

Accuracy is the closeness with which the instrument reading approaches the true value of the variable under measurement. Accuracy is determined as the maximum amount by which the result differs from the true value.

Accuracy of the measured signal depends upon the following factors:

- Intrinsic accuracy of the instrument itself;
- Accuracy of the observer; • Variation of the signal to be measured;
- Whether or not the quantity is being truly impressed upon the instrument

2. Precision:

Precision is a measure of the reproducibility of the measurements, i.e., precision is a measure of the degree to which successive measurements differ from one another.

3. Resolution:

If the input is slowly increased from some arbitrary value it will be noticed that the output does not change at all until the increment exceeds a certain value called the resolution or discrimination of the instrument.

Resolution is sometimes referred as sensitivity.

The largest change of input quantity for which there is no output of the instrument is called the dead zone of that instrument

The sensitivity gives the relation between the input signal to an instrument or a part of the instrument system and the output. Thus, the sensitivity is defined as the ratio of output signal or response of the instrument to a change of input signal or the quantity under measurement.

4. Speed of response: Quickness of the instrument

MEASUREMENT OF ERRORS:

It is impossible to measure the exact value of the measurand. There is always some difference between the measured value and the absolute or true value of the unknown quantity (measurand), which may be very small or may be large. The difference between the true or exact value and the measured value of the unknown quantity is known as the **absolute error of the measurement**.

If δA = absolute error of the measurement,

A_m and A = measured and absolute value of the unknown quantity ,

Then, δA may be expressed as

$$\delta A = A_m - A$$

Sometimes, δA is denoted by ϵ_0 .

The relative error is the ratio of absolute error to the true value of the unknown quantity to be measured,

$$\text{i.e., relative error, } \epsilon_r = \frac{\delta A}{A} = \frac{\epsilon_0}{A} = \frac{\text{Absolute error}}{\text{True value}}$$

When the absolute error $\epsilon_0 (= \delta A)$ is negligible, i.e., when the difference between the true value A and the measured value A_m of the unknown quantity is very small or negligible, then, the relative error may be expressed as,

$$\epsilon_r = \frac{\delta A}{A_m} = \frac{\epsilon_0}{A_m}$$

The relative error is generally expressed as a fraction, i.e., 5 parts in 1000 or in percentage value,

$$\text{i.e., percentage error} = \epsilon_r \times 100 = \frac{\epsilon_0}{A_m} \times 100$$

The measured value of the unknown quantity may be more than or less than the true value of the measurand. So, the manufacturers have to specify the deviations from the specified value of a particular quantity in order to enable the purchaser to make proper selection according to his requirements. The limits of these deviations from specified values are defined as limiting or guarantee errors.

magnitude of a given quantity having a specified magnitude A_m and a maximum or a limiting error $\pm \delta A$ must have a magnitude between the limits

$$A_m - \delta A \text{ and } A_m + \delta A$$

or, $A = A_m \pm \delta A$

For example, the measured value of a resistance of $100 \, \Omega$ has a limiting error of $\pm 0.5 \, \Omega$. Then the true value of the resistance is between the limits 100 ± 0.5 , i.e., 100.5 and $99.5 \, \Omega$.

TYPES OF ERRORS :

- Gross error
- Systematic error
- Random error